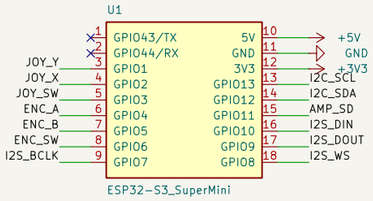
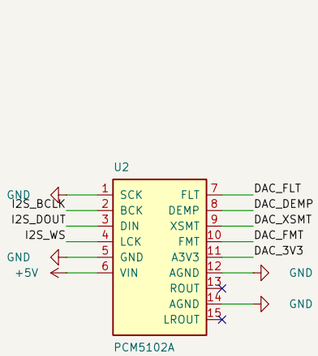
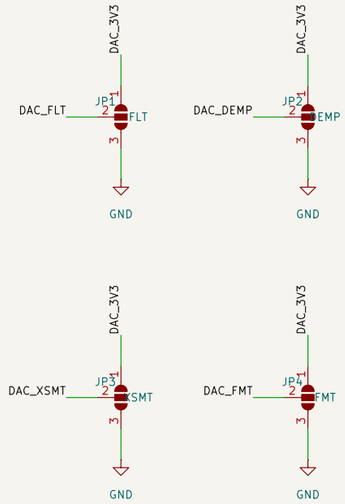
ESP32-S3 SuperMini



USB-C on the SuperMini powers everything: its 5V pin feeds the DAC (and amp), its 3.3 V regulator feeds the mic, IMU, joystick, OLED and encoder.

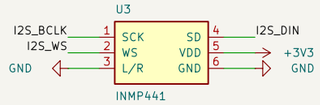
DAC (GY-PCM5102, line out)





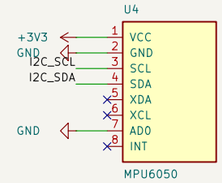
JP1-JP4 stand in for the H1L-H4L bridges on the back of the DAC board, out of reach once it is mounted. Leave them OPEN if those bridges are set. Otherwise bridge FLT=L, DEMP=L, XSMT=H (unmute), FMT=L. H is DAC_3V3, the module's own 3.3 V regulator output, never the board 3V3.

Microphone (INMP441, I2S)



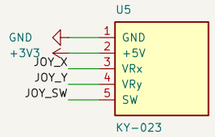
Sound hole UP: the metal can faces the carrier.
L/R to GND: data in the left slot.

Tilt (GY-521 / MPU-6050, I2C 0x68)



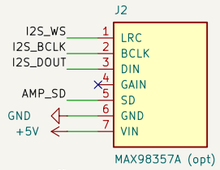
ADO to GND: address 0x68. The chip's +X axis points to the nose (the USB / jack edge).

Joystick (KY-023)



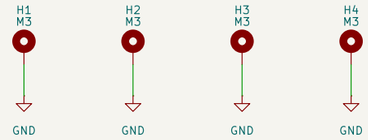
The "+5V" pin goes to 3V3: the ESP32 ADC reads 0-3.3 V. SW uses the internal pull-up.

Speaker amp header (optional MAX98357A)



GAIN floating = 9 dB. SD (GPIO11) low = amp off.
VIN from USB 5V.

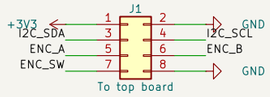
Mounting (M3, 11 mm standoffs)



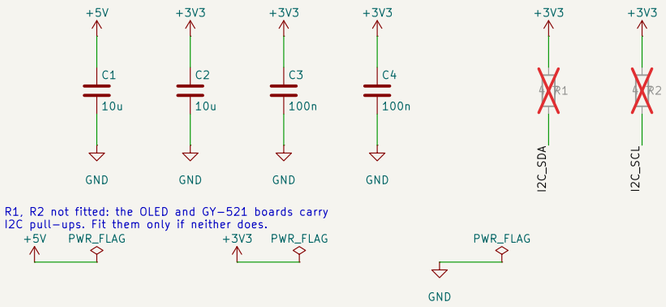
Holes tie both boards' ground pours through metal standoffs.

Stack: bottom board, 11 mm M3 standoffs, top board. The 2x4 socket (8.5 mm) and the top board's header (2.5 mm plastic) mate at exactly that height.

To the top board (2x4 socket, 8.5 mm)



Supply decoupling, I2C pull-ups



R1, R2 not fitted: the OLED and GY-521 boards carry I2C pull-ups. Fit them only if neither does.

Mates with samplooler-top through J1 (2x4, 11 mm stack)
ESP32-S3 SuperMini carrier: DAC, mic, IMU, joystick, amp header

Samplooler

Sheet: /
File: samplooler-bottom.kicad_sch

Title: Samplooler bottom board: modules

Size: A3 | Date: 2026-09-25 | **Rev: A**
KiCad E.D.A. 10.0.6 | Id: 1/1